Neural Nets and Deep Learning Aksel Kretsinger-Walters, adk2164
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COMS 4776: Fall 2025 **Due:** 11/21/2025 11:59 PM ET

(1) Written Problems: Batch Size vs. Learning Rate

(a) As batch size increases, how do you expect the optimal learning rate to change?

As batch size increases, I expect the optimal learning rate to also increase. This is because larger batch sizes provide a more accurate estimate of the gradient, allowing for larger steps to be taken in the parameter space without overshooting the minimum.

1.1 Training steps vs. batch size

(a) For the three points (A, B, B), in Figure 1, which one has the most efficient batch size. Point C has the most efficient batch size. Lets assume that point C is the batch size s.t. the compute is used quasi-optimally. This means that as batch size drastically decreases, the number of batches needed to cover the entire dataset surpasses the number of available compute units, leading to underutilization of resources (point A resides in this region of the chart - many more training steps are required). On the other hand, point B's batch sizes are so large that they exceed the capacity of the compute units, leading to inefficiencies due to memory constraints and increased communication overhead.

(b) Fill in the blanks.

Point A: Regime: Noise-dominated Potential acceleration: Increase batch size
Point B: Regime: Curvature-dominated Potential acceleration: Decrease batch size

(1.2) Model size, dataset size, and compute

(a) Best way to improve model **test** performance (C) Increase the model size. With more PF-days of compute, increasing the model size allows for capturing more complex patterns in the data, leading to better generalization and improved test performance.